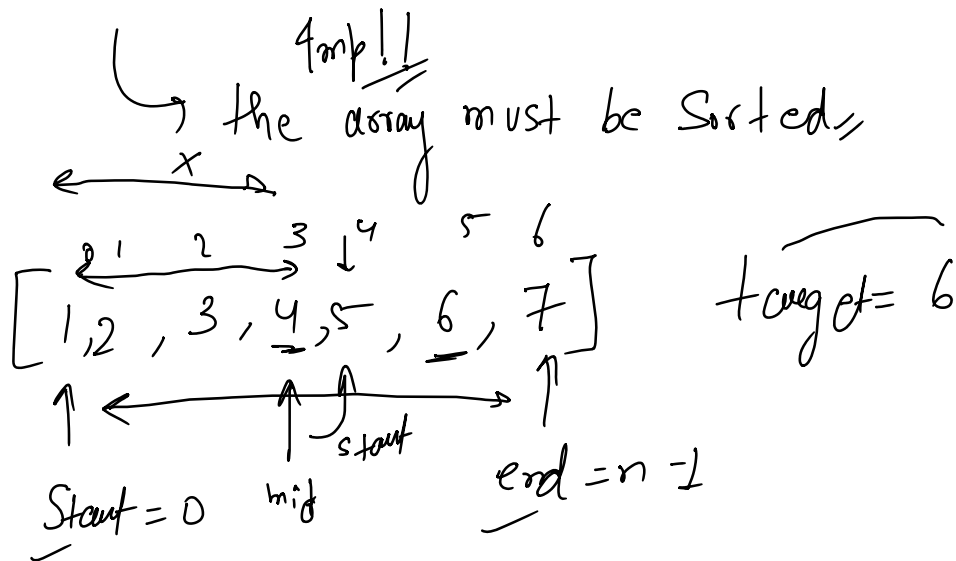


↳ Binary Search.

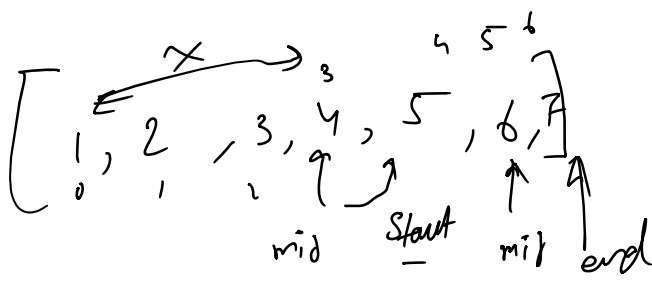


st

$$start = 0 \quad end = 6$$

$$mid = \frac{start + end}{2} = 3$$

$$arr[mid] == target$$



$$mid = \frac{start + end}{2}$$

$$\frac{4 + 6}{2} = 5$$

0 1 2 3 4 5 6

$$\frac{4+6}{2} \Rightarrow$$

```
int n = arr.length;
```

```
int start = 0, end = n - 1;
```

```
while (start <= end) {
```

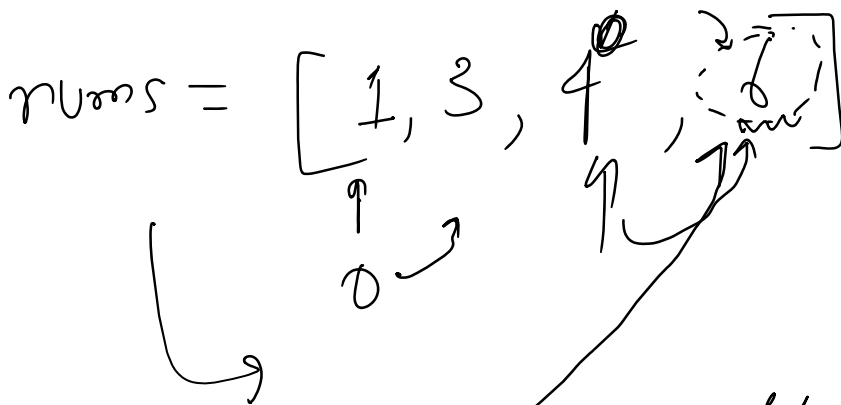
```
    int mid =  $\frac{start + end}{2}$ ;
```

```
    if (arr[mid] == target) return mid;
```

```
    else if (arr[mid] > target) end = mid - 1;
```

```
    else {
        start = mid + 1;
```

```
    }
    return -1;
```



target = 5

return i;

5